Machine Learning Capstone Project

Rain in Australia — Weather Prediction & Analysis

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Dataset	Weather Dataset - Rain in Australia (Kaggle)
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Google Colab Notebook Link

https://colab.research.google.com/drive/14tdWhMwQ6xcoBq6x-sjq6ODVzqNL9Kq2?usp=sharing

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Section I — Introduction

1.1 Project Goal

This project applies a full Machine Learning pipeline to the **Rain in Australia** weather dataset. Three distinct ML tasks are addressed:

- **Classification:** Predict whether it will rain the next day (RainTomorrow: Yes/No) using Logistic Regression, Decision Tree, and Random Forest.
- **Regression:** Predict the maximum temperature (MaxTemp) for a given day using Linear Regression.
- **Clustering:** Discover natural weather patterns across Australian cities using K-Means Clustering with PCA visualization.

1.2 Dataset Description

Source	Kaggle — Weather Dataset: Rain in Australia
URL	kaggle.com/datasets/jsphyg/weather-dataset-rattle-package
Rows	145,460 observations
Columns	23 features (16 numeric, 7 categorical)
Period	2007 – 2017 (10 years of daily weather)
Locations	49 weather stations across Australia
Target (Classification)	RainTomorrow — Binary (No: 110,316 / Yes: 31,877)
Target (Regression)	MaxTemp — Continuous (degrees Celsius)

Section II — Data Preprocessing & EDA

2.1 Missing Values Analysis

The dataset contained missing values across most features. Columns with extremely high missing rates were removed to avoid introducing noise into the models:

Sunshine	48.0%	Dropped
Evaporation	43.2%	Dropped
Cloud9am	38.4%	Dropped
Cloud3pm	40.8%	Dropped
Pressure9am	10.4%	Filled with Median
Pressure3pm	10.3%	Filled with Median
WindGustDir	7.1%	Filled with Mode
Other num.	< 3%	Filled with Median
Other cat.	< 3%	Filled with Mode

2.2 Feature Engineering & Encoding

- **Date Decomposition:** The Date column was parsed into Year, Month, and Day to capture temporal patterns such as seasonality.
- **Binary Encoding:** RainToday and RainTomorrow were mapped from Yes/No to 1/0.
- **One-Hot Encoding:** Categorical columns (Location, WindGustDir, WindDir9am, WindDir3pm) were encoded using get_dummies with drop_first=True to avoid multicollinearity.
- **Feature Scaling:** StandardScaler was applied to all features before training Logistic Regression and K-Means (distance-sensitive algorithms).

2.3 Key EDA Findings

- The target variable is **imbalanced**: 78% No Rain vs. 22% Rain — addressed by using stratify=y in the train/test split.
- Humidity3pm and Pressure3pm showed the strongest correlation with RainTomorrow.
- MaxTemp and MinTemp are highly correlated ($r \approx 0.97$), making MaxTemp a suitable and predictable regression target.
- The dataset spans 49 locations with varying climate profiles, motivating the clustering analysis.

2.4 Data Split

Train / Test Split	80% Training — 20% Testing
Random State	42 (reproducible)
Stratify	Yes — preserves class ratio in Classification task
Train Samples	~113,752 rows
Test Samples	~28,439 rows

Section III — Modeling and Results

3.1 Classification Task — Predicting RainTomorrow

Three classification models were trained to predict whether it will rain the following day. All models used the same 80/20 stratified split.

84.56%
Logistic Regression Accuracy

83.98%
Decision Tree Accuracy

85.48%
Random Forest Accuracy (Bonus)

Logistic Regression — Detailed Results:

No Rain (0)	0.87	0.95	0.90	22,064
Rain (1)	0.73	0.49	0.59	6,375
Weighted Avg	0.84	0.85	0.83	28,439

Confusion Matrix Analysis (Logistic Regression):

Actual: No	20,898 (TN)	1,166 (FP)
Actual: Yes	3,226 (FN)	3,149 (TP)

Precision vs. Recall Trade-off: The model achieves high recall (0.95) for No Rain cases but lower recall (0.49) for Rain. In a weather application, **minimizing False Negatives is critical** — predicting no rain when it actually rains is more costly than the reverse. Future improvement: applying class weights or SMOTE oversampling to improve Rain class recall.

3.2 Regression Task — Predicting MaxTemp

A Linear Regression model was trained to predict the maximum daily temperature (MaxTemp). RainTomorrow was excluded from features to prevent task overlap.

1.9745
MSE

1.4052°C
RMSE

0.9609
R^2 Score

Interpretation:

• **R^2 = 0.9609** — The model explains 96.09% of the variance in MaxTemp, indicating an excellent fit.

• **RMSE = 1.41°C** — On average, predictions deviate by only 1.41 degrees Celsius from actual values, which is highly accurate for weather prediction.

• The Actual vs. Predicted plot confirms predictions closely follow the ideal diagonal line, with no major systematic bias.

• The residuals distribution is approximately normal and centered at zero, confirming that Linear Regression assumptions are satisfied.

3.3 Clustering Task — K-Means Weather Patterns

K-Means clustering was applied to the full feature set (excluding RainTomorrow) after StandardScaler normalization, to discover natural weather groupings.

Optimal K Selection — Elbow Method:

Inertia was computed for K = 1 to 10. The elbow plot showed a sharp drop between K=1 and K=3, with diminishing returns beyond K=3. **K=3 was selected as the optimal number of clusters.**

~0.20+
Silhouette Score (K=3)

3
Optimal K (Elbow Method)

All (excl. target)
Features Used for Clustering

Cluster Interpretation:

Cluster 0	Hot & Dry	High MaxTemp, Low Humidity, Low Rainfall
Cluster 1	Mild	Moderate Temperatures, Moderate Humidity
Cluster 2	Cool & Wet	Low MaxTemp, High Humidity, Higher Rainfall

PCA was applied (2D and 3D) to visualize high-dimensional clusters. The 2D PCA plot shows reasonably separated cluster regions, confirming the K-Means groupings capture meaningful patterns in the data.

Section IV — Conclusion

4.1 Best-Performing Model per Task

Classification	Random Forest (Bonus)	Accuracy	85.48%
Regression	Linear Regression	R^2 Score	0.9609
Clustering	K-Means (K=3)	Silhouette	~0.20+

4.2 Challenges Encountered

• **Class Imbalance:** The 78/22 split between No Rain / Rain caused the classifier to be biased toward the majority class. This was partially mitigated by stratified splitting but remains a future improvement area.

• **High Dimensionality after Encoding:** One-hot encoding of Location and wind direction columns significantly expanded the feature space. PCA helped with visualization but was not applied to model training.

• **Silhouette Score for Clustering:** A score around 0.20 indicates moderate cluster separation — expected given the high-dimensional and continuous nature of weather data.

4.3 Potential Next Steps

• Apply **SMOTE** or class weighting to improve Rain class recall in classification.

• Try **Gradient Boosting (XGBoost / LightGBM)** for potentially higher accuracy.

• Apply **PCA before model training** to reduce dimensionality and training time.

• Use **time-series cross-validation** to respect the temporal structure of weather data.

• Explore **DBSCAN or Hierarchical Clustering** as alternatives to K-Means.